

Experimental Dynamics and Data Processing

Laboratory 4 – FRF Random

Useful MATLAB Functions

1. Data selection / loading

Usage	Main functions	Similar functions
File selection	uigetfile	uigetdir
Load data	load	importdata, readmatrix, readtable
Path management	cd	fullfile, pwd, dir

2. Signal calibration and time vector

Usage	Main functions	Similar functions
Signal length	length	size, numel
Create time vector	:	linspace
Column selection	indexing	–

3. Plotting the signals

Usage	Main functions	Similar functions
Create a new figure	figure	–
Multiple plots in one figure	subplot	tiledlayout, nexttile
Plot a signal	plot	–
Log Scale	semilogy	semilogx
Axis labels	xlabel, ylabel	–
Add title	title	sgtitle
Legend	legend	–
Grid	grid on	grid minor
Linked axes	linkaxes	–
Axis formatting	set	gca
Second y-axis	yyaxis	–

4. Local function creation

Usage	Main functions	Similar functions
Define a function	<code>function</code>	–
Call function with multiple outputs	<code>[out1, out2] = myFun(inputs)</code>	–
Suggested functions to create	normalised FFT, averaged power/cross spectrum, etc	–

5. FFT, frequency vector and FRF

Usage	Main functions	Similar functions
Fast Fourier Transform	<code>fft</code>	<code>ifft</code> (inverse <code>fft</code>)
Complex conjugate	<code>conj</code>	–
Modulus of complex number	<code>abs</code>	–
Phase angle	<code>angle</code>	<code>unwrap</code>
Even/odd check	<code>mod</code>	<code>floor</code>
Window	<code>hanning</code>	<code>hamming</code> , <code>blackman</code>
Element-wise complex division	<code>./</code>	<code>rdivide</code>
Compute FFT input/output	<code>fft</code> (with window)	–
FRF	$H=Y/X$	–

6. Power spectra and cross-spectra (averaged)

Usage	Main functions	Similar functions
Number of points in window	<code>round</code>	<code>floor</code>
Number of overlapping points	<code>floor</code>	–
Number of sub-records	<code>fix</code>	–
Compute FFT of Windowed Signal	<code>fft</code>	
Power/Cross-spectrum (one record) (see presentation)	arithmetic operations	local functions
Average over records	<code>mean</code>	<code>sum</code>
While loop (sub-record sliding)	<code>while</code>	<code>for</code>
Matrix pre-allocation	<code>zeros</code>	–

7. Coherence function

Usage	Main functions	Similar functions
Coherence formula (see presentation)	arithmetic operations	local function
Element-wise power	.^	power
Element-wise multiplication	.*	

8. H1 and H2 FRF estimators

Usage	Main functions	Similar functions
H1, H2 formula (see presentation)	arithmetic operations	local function
Overlay H1 and H2	semilogy, hold on	–

9. Display and formatting

Usage	Main functions	Similar functions
Display message	disp	fprintf
Convert number to string	num2str	sprintf, string
Axis formatting	set	–
Axis limits	axis tight	xlim, ylim

10. Comparison and analysis

Usage	Main functions	Similar functions
Overlay two curves	hold on / hold off	–
Loop over window lengths	for	–
Conditional branching	if/else	–